WORKED SOLUTION

STEP II 1999, Question 11

Mechanics • Projectiles and Moving Targets

STEP II 1999, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q11

The key idea

The target stays on the circle, so the shell must land on the horizontal plane at distance exactly R from O; and the gun can always be swung round to face wherever the target will be. So everything comes down to two facts about one projectile: its range and its time of flight. The rotation angle is $\beta = \omega T$, and eliminating α with $\sin^2\alpha + \cos^2\alpha = 1$ produces the quartic. The last part is then Vieta's formulas plus a chord of a circle.

Solution

We model the shell as a particle moving under gravity alone (no air resistance), fired from O on the horizontal plane containing the circle, with $0 < \alpha < \frac{1}{2}\pi$.

Part 1: when $V^2 < gR$

Take axes along the ground (in the plane of firing) and vertically up. At time t,

$$x = Vt\cos\alpha, \qquad y = Vt\sin\alpha - \tfrac{1}{2}gt^2.$$

The shell returns to the plane ($y = 0$, $t > 0$) at

$$T = \frac{2V\sin\alpha}{g}, \qquad \text{at horizontal distance} \qquad x(T) = \frac{2V^2\sin\alpha\cos\alpha}{g} = \frac{V^2\sin 2\alpha}{g}.$$

Since $\sin 2\alpha \le 1$, the shell lands no further than V^2/g from O, whatever the vertical plane. The target is always on the plane at distance R from O, so a hit needs $R \le V^2/g$, that is $V^2 \ge gR$. Hence if $V^2 < gR$ the shell can never hit the target, as required.

Part 2: the equation for β

Now $V^2 > gR$. The shell hits the target at time T and distance R, so

$$T = \frac{2V\sin\alpha}{g}, \qquad R = \frac{2V^2\sin\alpha\cos\alpha}{g}.$$

The target turns at constant rate ω, so $\beta = \omega T$, which gives

$$\sin\alpha = \frac{g\beta}{2V\omega}.$$

Substituting this into the range equation,

$$\cos\alpha = \frac{gR}{2V^2\sin\alpha} = \frac{gR}{2V^2}\cdot\frac{2V\omega}{g\beta} = \frac{R\omega}{V\beta}.$$

Now $\sin^2\alpha + \cos^2\alpha = 1$ gives

$$\frac{g^2\beta^2}{4V^2\omega^2} + \frac{R^2\omega^2}{V^2\beta^2} = 1,$$

and multiplying through by $4V^2\omega^2\beta^2$,

$$g^2\beta^4 - 4\omega^2V^2\beta^2 + 4R^2\omega^4 = 0,$$

which is the required equation.

Ask yourself Why are there exactly two values, and not up to four, or none?

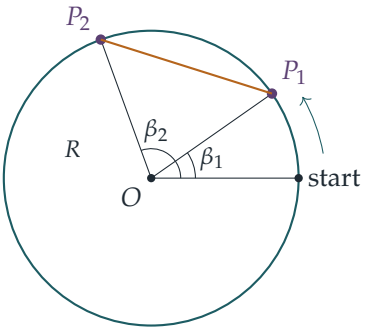
As a quadratic in β^2 its discriminant is

$$16\omega^4V^4 - 16g^2R^2\omega^4 = 16\omega^4(V^4 - g^2R^2) > 0,$$

since $V^2 > gR$. So there are two distinct real values of β^2. Their sum $4\omega^2V^2/g^2$ and product $4R^2\omega^4/g^2$ are both positive, so both values of β^2 are positive. The angle $\beta = \omega T$ is positive, so each value of β^2 gives exactly one β, and we have exactly two candidates $\beta_1 < \beta_2$.

Each one really does give a hit. For either root, the numbers $g\beta/(2V\omega)$ and $R\omega/(V\beta)$ are positive and (by the equation) their squares add up to 1, so there is an $\alpha \in (0, \frac{1}{2}\pi)$ with these as its sine and cosine. Firing at that elevation, in the vertical plane through the point the target will reach after turning through β, gives a time of flight β/ω and a range R, so the shell meets the target. Hence there are exactly two possible values of β. (They correspond to the two complementary elevations α and $\frac{1}{2}\pi - \alpha$ that give range R.)

Part 3: the distance P_1P_2



Both points of impact are on the circle, measured anticlockwise from the target's position at the moment of firing, at angles β_1 and β_2. So the angle P_1OP_2 is $\beta_2 - \beta_1$ (reduced modulo 2π if necessary, which does not affect the chord), and the isosceles triangle OP_1P_2 gives

$$P_1P_2 = 2R\left|\sin\frac{\beta_2-\beta_1}{2}\right|.$$

From the quadratic in β^2,

$$\beta_1^2+\beta_2^2 = \frac{4\omega^2V^2}{g^2}, \qquad \beta_1^2\beta_2^2 = \frac{4R^2\omega^4}{g^2},$$

and since $\beta_1, \beta_2 > 0$ we can take the positive square root: $\beta_1\beta_2 = 2R\omega^2/g$. Therefore

$$\begin{aligned}(\beta_2-\beta_1)^2 &= \beta_1^2+\beta_2^2-2\beta_1\beta_2\\ &= \frac{4\omega^2V^2}{g^2} - \frac{4R\omega^2}{g} = \frac{4\omega^2}{g^2}\left(V^2-Rg\right),\end{aligned}$$

so $\beta_2-\beta_1 = \dfrac{2\omega}{g}\sqrt{V^2-Rg}$. Hence

$$P_1P_2 = 2R\sin\left(\frac{\omega}{g}\sqrt{V^2-Rg}\right),$$

as required. (Strictly this needs the sine to be non-negative, that is $\frac{\omega}{g}\sqrt{V^2-Rg} \le \pi$; otherwise the distance is the modulus of this expression.)

Checks

- As $V^2 \to gR^+$ the two elevations both tend to 45°, so P_1 and P_2 should merge: indeed the formula tends to 0.
- Check $\beta_1\beta_2 = 2R\omega^2/g$ directly: $\beta_1\beta_2 = (2V\omega/g)^2\sin\alpha_1\sin\alpha_2$, and with $\alpha_2 = \frac{1}{2}\pi - \alpha_1$ this is $(2V\omega/g)^2\sin\alpha_1\cos\alpha_1 = (4V^2\omega^2/g^2)(gR/2V^2)$, as expected.
- Numerically, with $g = 9.8$, $R = 10$, $V = 14$, $\omega = 0.3$ (SI units), stepping both trajectories gives a chord of about 5.97 m, matching $20\sin\left(0.3\sqrt{98}/9.8\right) \approx 5.969$.

Common slips

- Forgetting that the gun can be aimed in any vertical plane: the first part is about range alone, not about where the target happens to be.
- Taking the square root of $\beta_1^2\beta_2^2$ without saying why $\beta_1\beta_2$ is positive.
- Writing $P_1P_2 = R(\beta_2-\beta_1)$, which is the arc length, not the straight-line distance.

Beyond STEP

Hitting a moving target is a small *pursuit* or *interception* problem: you solve for the time of impact first, then for the aim. Eliminating a parameter between two equations to get a polynomial, and then reading off what you need from its coefficients without solving it, is a technique you will use constantly at university. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats projectiles and rotating motion with this kind of care.

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